HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION COMMUNICATION TECHNOLOGY

# CAPSTONE PROJECT REPORT:

# ADVENTURE GAME

**Supervised by:**

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# WORK DISTRIBUTION

| No. | Name | ID | Task | Process |
|---|---|---|---|---|
| 1 | Bui Khac Chien | 20220059 | - Character<br>- System Design<br>- Enemy and Boss Movement<br>- Video | 100% |
| 2 | Nguyen Tuan Cuong | 20214950 | - Effect<br>- Audio<br>- View<br>- Input Handler | 100% |
| 3 | Dau Van Can | 20214879 | - Map<br>- Item System | 100% |
| 4 | Vu Tung Linh | 20210523 | - ECS Engines<br>- Presentation | 100% |
| 5 | Nguyen Phuong Uyen | 20204933 | - Menu Options<br>- UI<br>- Powerpoint | 100% |

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## 1. Problem Statement

In the current landscape of gaming, many games focus on action and adventure elements but lack the integration of structured, efficient development practices. The challenge was to create a game that not only entertains players but also showcases the power of object-oriented programming in game development. The project aimed to develop a 2D adventure game with a focus on monster slaying using the Java programming language and the libgdx framework. This required addressing several key challenges, such as designing the game mechanics, creating engaging gameplay elements, and ensuring smooth performance across different platforms. By doing so, the group sought to demonstrate how object-oriented principles could be applied effectively in game development

## 2. Content

### 2.1. Tools Used

*Table 1: Tool Used*

| | |
|---|---|
| **Programming Language** | Java |
| **Development tools** | IntelliJ IDEA IDE, Gradle |
| **Framwork** | Libgdx |
| **Library** | Libgdx game development tools such as TexturePacker, Skin Composer, etc. |

### 2.2. Tasks Performed

**Steps in Game Development**

- Design and implement the game system.
- Retrieve data from the database.
- Divide the screens for different functions (menu, dead screen, etc.).
- Design the map and implement it in the system.
- Create an ECSEngine for objects and their interaction systems:
    - Attack system.
    - Movement system.
    - Animation display system.
- Design extensions (UI, sound, etc.).
- Complete and demonstrate the product.

### 2.3. UML Diagram of the Program

Below is an image of the UML diagram for the program.

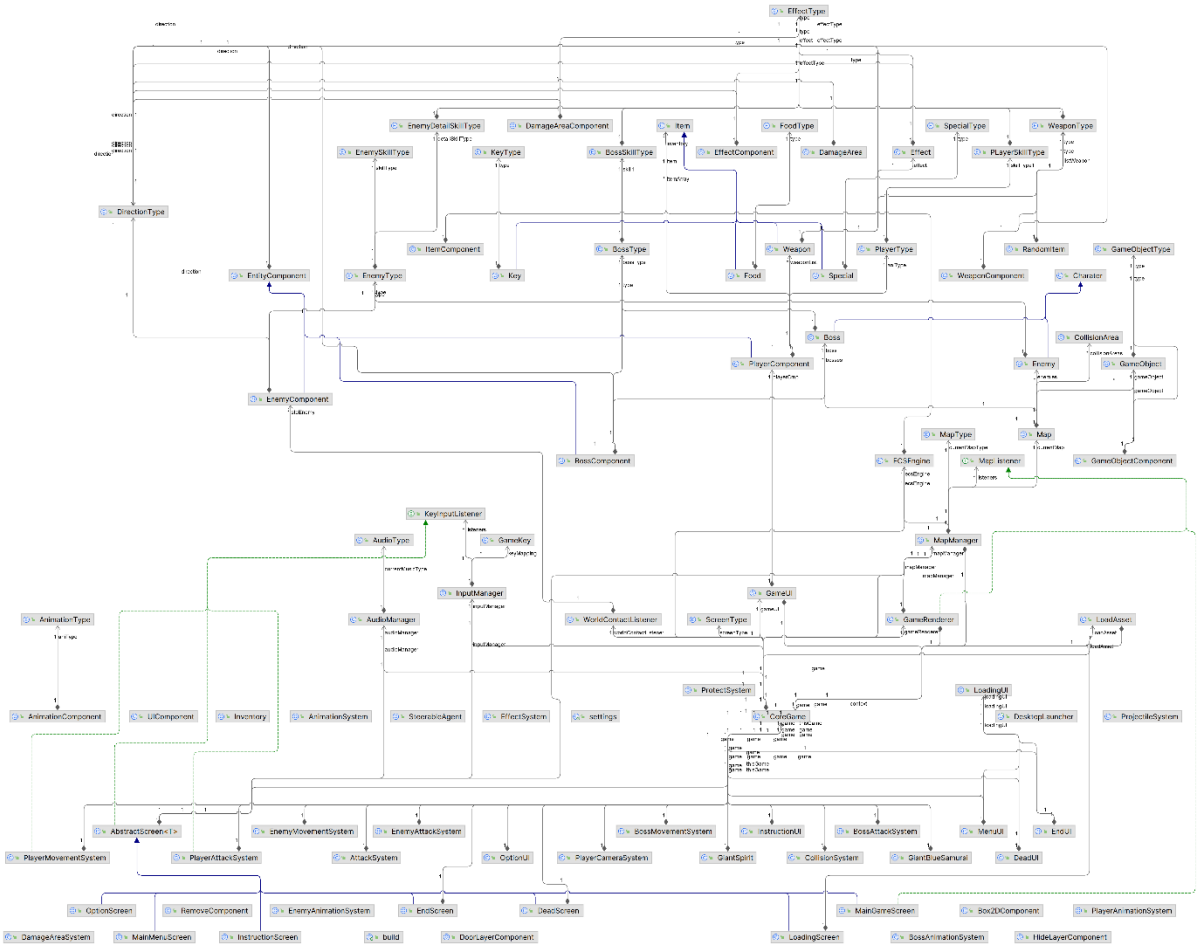



*Figure 1: UML for Adventure Game*

## 3. Result

Here are some images of the project:

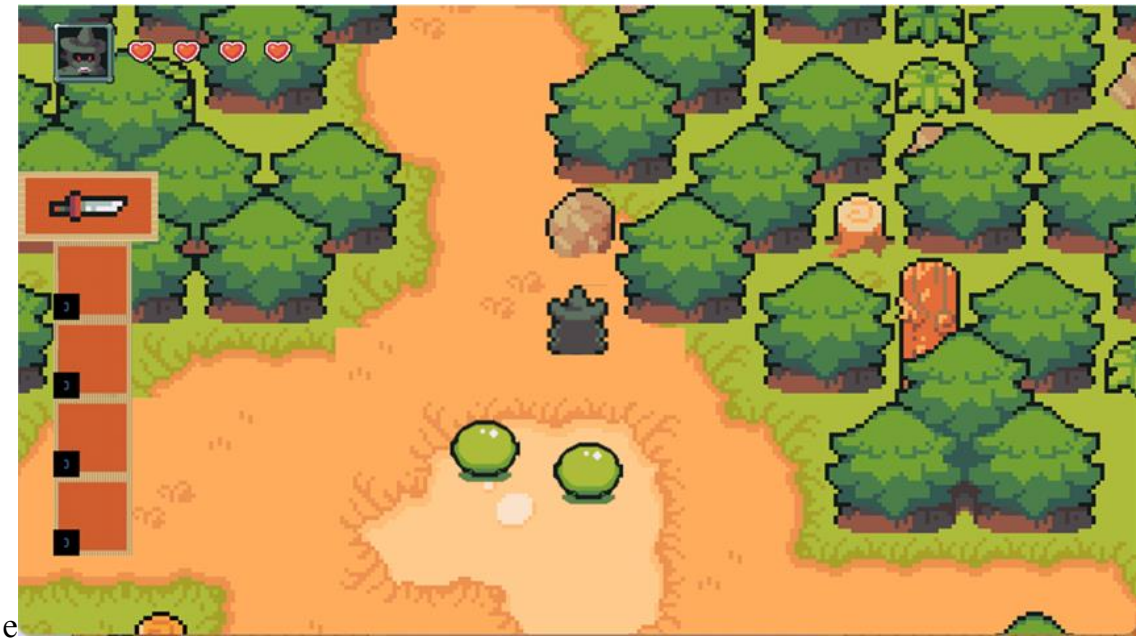
e 

*Figure 2: Starting Map*

*Figure 3: Boss Battle Map*

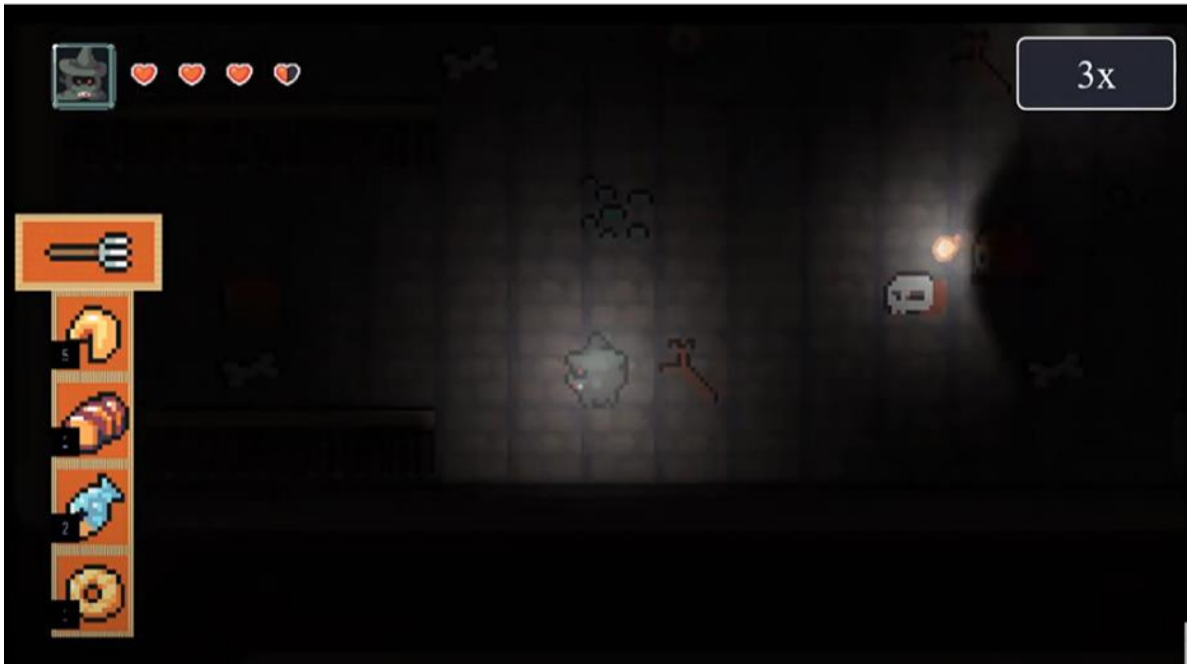



*Figure 4: Item Search Map*

*Figure 5: Adventure Map*

## 4. Conclusion

Through the process of implementation, Group 63 has effectively applied the principles of object-oriented programming to build the structure and gameplay mechanics for the game.

## 5. Authors

Authors: Bui Khac Chien, Nguyen Tuan Cuong, Dau Van Can, Vu Tung Linh, Nguyen Phuong Uyen

**[Source]:** https://github.com/RemMyDream/Adventure_Game.git
**[Documentation]:** https://libgdx.com/
**[Video]:** Link